

Mathematical Analysis Lecture 2 | 定积分.

e.g. 求 $y = \sin x$ 与 x 轴在 $[0, \pi]$ 上的面积 S

$$S = \int_0^{\pi} \sin x \, dx = -\cos x \Big|_0^{\pi} = 2.$$

e.g. $\int_0^2 \frac{(x-1)^2+1}{(x-1)^2+x^2(x-1)^2} dx = \int_0^2 d \arctan \left(\frac{x(x-2)}{x-1} \right) = \arctan \frac{x(x-2)}{x-1} \Big|_0^2 = 0?$

已知: $\frac{(x-1)^2+1}{(x-1)^2+x^2(x-1)^2} > 0$, 由于积为得 $\arctan \frac{x(x-2)}{x-1}$ 在 $x=1$ 处无定义: 不能用 Newton-Lebniz 积分.

正解:
$$F(x) = \begin{cases} \arctan \frac{x(x-2)}{x-1} & x \in [0, 1) \\ \frac{\pi}{2} & x = 1 \end{cases} \quad \tilde{F}(x) = \begin{cases} \arctan \frac{x(x-2)}{x-1} & x \in (1, 2] \\ -\frac{\pi}{2} & x = 1 \end{cases}$$

$$\therefore \int_0^2 \frac{(x-1)^2+1}{(x-1)^2+x^2(x-1)^2} dx = \arctan \frac{x(x-2)}{x-1} \Big|_0^1 + \arctan \frac{x(x-2)}{x-1} \Big|_1^2 = \pi.$$

改变有限个点, 乃至可数无穷个点, 不改变测度与积分结果.

定积分的分部积分法:

设 $u(x), v(x) \in C^1[a, b]$ (一阶导存在且连续) 则 $\int_a^b u(x) v'(x) dx = u(x) v(x) \Big|_a^b - \int_a^b u'(x) v(x) dx$.

证明: $\because (uv)' = u'v + uv'$ 两侧定积分: QED

e.g. $\int_0^{\frac{1}{2}} \arcsin x \, dx = x \arcsin x \Big|_0^{\frac{1}{2}} + \frac{1}{2} \int_0^{\frac{1}{2}} (1-x^2)^{-\frac{1}{2}} d(1-x^2) = \frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}.$

定积分的换元积分法: (第一类换元法).

设 $f(x) \in C[a, b]$, $x = \varphi(t)$ 满足: 1) $\varphi(t) \in C'[\alpha, \beta]$, $\varphi(\alpha) = a$, $\varphi(\beta) = b$.

2) 在 $[\alpha, \beta]$ 上, $\varphi(t) \in [a, b]$, 则 $\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(\varphi(t)) \cdot \varphi'(t) dt$.

证: 设 $F(x)$ 是 $f(x)$ 的一个原函数 则 $F(\varphi(t))$ 是 $f(\varphi(t)) \varphi'(t)$ 的原函数

$$\text{Thus: } \int_a^b f(x) dx = F(b) - F(a) = F(\varphi(\beta)) - F(\varphi(\alpha)) = \int_{\alpha}^{\beta} f(\varphi(t)) \varphi'(t) dt$$

e.g. $\int_0^a \sqrt{a^2 - x^2} dx \ (a > 0), \quad \text{令 } x = a \sin t, \quad \int_0^a \sqrt{a^2 - x^2} = \int_0^{\frac{\pi}{2}} |a \cos t| d(a \sin t) = \int_0^{\frac{\pi}{2}} a^2 \cos^2 t dt$
 $= a^2 \int_0^{\frac{\pi}{2}} \left(\frac{1}{2} \cos 2t + \frac{1}{2} \right) dt = \frac{\pi}{4} a^2. \quad (\text{其实就是 } \frac{1}{4} \text{ 个圆})$

e.g. $\int_0^4 \frac{x+2}{\sqrt{2x+1}} dx \quad \text{令 } t = \sqrt{2x+1}, \quad x = \frac{t^2-1}{2}, \quad \alpha = 1, \quad \beta = 3.$

$$\therefore \text{原式} = \int_1^3 \frac{\frac{1}{2}t^2 + \frac{3}{2}}{t} t dt = \int_1^3 \left(\frac{1}{2}t^2 + \frac{3}{2} \right) dt = \frac{22}{3}.$$

Inference: 设 $f(x) \in [-a, a]$, 若 $f(-x) = f(x)$, 则 $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.

若 $f(-x) = -f(x)$, 则 $\int_{-a}^a f(x) dx = 0$.

Inference: 设 $f(x)$ 是连续的周期函数, 周期为 T , 证明 $\int_a^{a+T} f(x) dx = \int_0^T f(x) dx$.

$$\text{记 } \Phi(a) = \int_a^{a+T} f(x) dx, \quad \Phi'(a) = F'(a+T) - F'(a) = f(a+T) - f(a) = 0.$$

$\therefore \Phi(a)$ 为常函数, $\Phi(a) = \Phi(0), \quad \text{QED}$

e.g. $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx$. 令 $t = \frac{\pi}{2} - x$ 相等性可知.

$$\text{求证: } I_n = \begin{cases} \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{\pi}{4} & \text{若 } n \text{ 为偶数} \\ \frac{n-1}{n} \cdot \frac{n-3}{n-2} \cdots \frac{4}{5} \cdot \frac{2}{3} & \text{若 } n \text{ 为奇数} \end{cases}$$

$$\text{证: 令 } u = \sin^{n-1} x, \quad v' = \sin x \quad \therefore I_n = \left[-\cos x \sin^{n-1} x \right]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-1} x \cos^2 x dx$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n \quad \therefore I_n = \frac{n-1}{n} I_{n-2}. \quad \text{即为所求.}$$